94 - SmartCare AI: Multimodal Fall Detection and Next-Day Fall Risk Prediction Prototype for Elderly Care

Client: Gary Wang

Allowed groups: 3 groups

Project background & goals

We are developing an AI-enabled elderly care monitoring system that aims to move fall management from reactive fall alarms to earlier risk identification. The current product concept combines a chest-worn 6-axis inertial measurement unit with sleep and activity information from a bedside monitoring hub. The system is designed for non-camera, privacy-preserving monitoring in home care and aged-care settings.

The goal of this capstone project is to design and implement a software prototype that can ingest wearable motion data and sleep-summary data, detect possible fall events, and estimate a next-day fall-risk level for an older adult. The prototype should output a daily risk score, a simple risk category such as low, medium, or high, and a short explanation of the key contributing factors, such as unstable movement patterns, reduced activity, or poor recent sleep.

Project scope

The project scope is to build a working software/system prototype over the Term 2 capstone period. The expected scope includes:

A data ingestion pipeline for 6-axis IMU data, including 3-axis accelerometer and 3-axis gyroscope signals, plus nightly sleep summary data.
Data preprocessing, timestamp alignment, signal-quality checks, wear-time or missing-data detection, and segmentation into useful activity windows.
A fall-detection prototype that classifies motion windows into fall-like events versus normal daily activities using public datasets and/or de-identified or simulated SmartCare sample data.
A next-day fall-risk prediction prototype that combines daily movement features and recent sleep-history features.
A simple backend API and dashboard that allows a user to upload or simulate data, view detected events, daily risk level, model confidence, and explanation messages.
An evaluation report comparing baseline and improved approaches, including motion-only, sleep-only, and multimodal models where feasible.

Out of scope: production mobile app deployment, real emergency phone calls, regulatory certification, clinical validation, hardware manufacturing, and medical advice. The final system should be a demonstrable prototype with clear documentation and a pathway for future development.

Project requirements

Data pipeline

Import IMU data in CSV or JSON format, including acceleration and gyroscope channels.
Import sleep summary data, such as sleep duration, sleep quality score, wake periods, or sleep-stage summary

where available.
Perform timestamp alignment, missing-data checks, outlier detection, and basic signal-quality assessment.
Segment motion data into windows suitable for fall detection, gait/activity analysis, and daily feature aggregation.

Fall detection model

Build a baseline fall-detection classifier using public fall datasets and/or SmartCare-provided simulated or de-identified data.
Explore traditional feature-based methods such as random forest, gradient boosting, or SVM, and optionally deep learning models such as 1D CNN, GRU, temporal CNN, or transformer-lite models.
Classify fall-like events versus normal activities of daily living.
Report model performance using accuracy, sensitivity/recall, specificity, precision, F1-score, and confusion matrix.
Include a thresholding or confidence mechanism to reduce excessive false alarms.

Next-day fall-risk prediction

Extract interpretable daily motion features, such as activity level, inactivity duration, movement variability, turning stability proxy, posture-transition instability proxy, and morning-versus-evening activity change.
Combine movement features with rolling sleep-history features, such as previous 3-day and 7-day sleep averages or trend changes.
Output a daily risk probability and map it to low, medium, or high risk.
Provide simple explanations such as “reduced activity compared with baseline”, “higher motion variability”, or “poor sleep trend over recent nights”.
Compare motion-only, sleep-only, and multimodal approaches if time permits.

System prototype

Develop a simple web dashboard or web application for demonstration.
Provide API endpoints for data upload, model inference, event history, and daily risk summary.
Visualize raw or processed sensor trends, detected fall events, risk score, risk category, and explanation.
Implement a simulated alert workflow based on event states such as suspected event, confirmed event, and cancelled event. This should be a software simulation only, not a real emergency-calling function.

Testing and documentation

Provide unit tests or integration tests for key data-processing and API functions.
Provide clear documentation for setup, dataset format, model training, inference, and dashboard usage.
Provide a final technical report explaining design decisions, model performance, limitations, and recommended next steps.

Required knowledge and skills

Students should ideally have knowledge or interest in:

Python programming
Machine learning and data preprocessing
Time-series or sensor-data analysis
Basic deep learning using PyTorch, TensorFlow, or similar frameworks
scikit-learn or equivalent machine-learning libraries
Backend development using FastAPI, Flask, Node.js, or similar

Web application development using React, Vue, or a comparable framework
Data visualization and dashboard design
Basic database usage, such as SQLite, PostgreSQL, or MongoDB
Git version control and software testing
Basic understanding of privacy, responsible AI, and healthcare-adjacent software limitations

Optional but useful:

IoT or wearable-sensor experience
Signal processing
Human-computer interaction
Cloud deployment experience
Experience with ONNX or lightweight model deployment

Expected outcomes and deliverables

Source code repository with clear project structure.
Data ingestion and preprocessing pipeline for IMU and sleep-summary data.
Baseline fall-detection model and evaluation results.
Next-day fall-risk prediction model producing low/medium/high risk categories.
Basic explainability module showing key risk contributors.
Web dashboard or web application demo.
Backend API for data upload, inference, and result retrieval.
Model evaluation report, including metrics and limitations.
User guide for running the prototype and using the dashboard.
Developer documentation, including environment setup, data format, model training, and future improvement suggestions.
Final presentation and demo video.
Handover package suitable for SmartCare/Syner Global's internal technical team to review and continue development.

Disciplines related to the project

Software Development;Web Application Development;Mobile Application Development;Computer Science and Algorithms;Artificial Intelligence (Machine/Deep Learning, NLP);Human Computer Interaction (HCI);Internet of Things (IoTs);Robotics;Computer Vision;Big data Analytics and Visualization;Cloud Computing;System/game Development

Resources provided

can provide the following resources:

Product requirement brief and system concept documentation.
Example or simulated 6-axis IMU datasets from chest-worn wearable scenarios.
Guidance on relevant public fall-detection datasets for pre-training and benchmarking.
Example sleep-summary data, either simulated or de-identified where appropriate.
Domain knowledge on elderly care workflows, fall-alert workflow, and caregiver requirements.
Weekly or fortnightly client feedback meetings.
Access to a sandbox cloud environment or API specification if required.

Optional access to sample hardware or recorded sensor logs, depending on availability and UNSW data/privacy requirements.
Clear constraints on privacy, non-medical positioning, and responsible use.

No personally identifiable health information will be required for the capstone.